

To,

Dated: Sept 19, 2026

Shri Muktanand Agarwal

Joint Secretary (Plant Protection),
Ministry of Agriculture and Farmers Welfare,
Government of India,
Krishi Bhawan, New Delhi.

Subject: Suggestions on the Draft “Banning of Carbosulfan Order, 2026” (S.O. 4649(E) dated 21 August 2026) – Request of leading Plant Protection Societies of India for scientific review and assessment of alternatives before prohibition

Ref: Draft “Banning of Carbosulfan Order, 2026”, S.O. 4649(E), dated 21 August 2026, published in the Gazette of India on 24 August 2026.

Sir,

With reference to the above Draft Order proposing a complete prohibition on the manufacture, import, transport, distribution, sale, and use of carbosulfan, we are submitting this joint representation on behalf of five premier national professional societies having a core mandate in plant protection across India:

1. Indian Phytopathological Society (IPS), New Delhi
2. Entomological Society of India (ESI), New Delhi
3. Nematological Society of India (NSI), New Delhi
4. Plant Protection Association of India (PPAI), Hyderabad

Following a joint national brainstorming session on “Carbosulfan in Indian Agriculture: Scientific Evidence, Agricultural Utility, Human Health, Environmental Safety and Options for a Rational Regulatory Transition” held on 19 September 2026 in Siddharth Hotel, New Delhi, the four national societies collectively arrived at a common understanding and opinion on the draft ban proposal. As institutional custodians of plant protection science in India, we respectfully submit the following evidence-based, farmer-sensitive consensus recommendations for your consideration before a final regulatory decision is enacted.

While we fully recognise the toxicological concerns associated with carbosulfan, particularly its anticholinesterase activity, acute toxicity, metabolic conversion to carbofuran and potential effects on non-target organisms. These concerns merit serious regulatory consideration. At the same time, a regulatory decision should distinguish between intrinsic hazard and actual exposure risk under approved agricultural use, and should consider the agricultural utility of the molecule, exposure pathways, availability and adequacy of alternatives, and the consequences of its withdrawal under Indian conditions. The available evidence also indicates that the Sri Lankan poisoning experience relates principally to intentional ingestion, often involving concentrated formulations, and should not by itself be directly extrapolated to occupational, dietary or other exposure arising from regulated agricultural use in India.

1. Human-health concerns deserve consideration, but risk should be assessed comprehensively

Carbosulfan is an intrinsically hazardous carbamate insecticide. The WHO classification places the technical active ingredient in Class II, moderately hazardous, while the Indian CIB&RC classification identifies it separately as highly toxic. Its anticholinesterase activity and conversion to carbofuran and other metabolites provide a legitimate basis for careful assessment of human-health risks.

The Sri Lankan experience has also demonstrated that intentional ingestion of carbosulfan, particularly in concentrated formulations, can result in substantial mortality. However, such evidence primarily concerns deliberate self-poisoning and should not automatically be equated with the risk arising from its regulated agricultural use. The final regulatory decision would therefore be strengthened by an explicit assessment of hazard, exposure and risk under Indian conditions, including occupational exposure, accidental and intentional poisoning, dietary exposure under Good Agricultural Practices, residues and the effectiveness of existing risk-mitigation measures.

2. Earlier scientific reviews recommended continuation, what new evidence warrants a different conclusion?

The proposed prohibition represents a significant departure from the conclusions of earlier Government-appointed expert reviews. The C.D. Mayee Expert Group specifically recommended that “Use of Carbosulfan be continued.” The Anupam Varma Expert Committee, which reviewed pesticides banned or restricted in other countries but registered in India, also placed carbosulfan in the category recommended for continued use. These earlier recommendations are not necessarily determinative of the present decision, since scientific evidence and regulatory considerations can evolve. However, for transparency and scientific consistency, it is important to establish what new evidence has emerged since these reviews-particularly in relation to toxicology, occupational exposure, dietary residues, environmental fate, field exposure or actual poisoning under Indian conditions, that now warrants a shift from continued use to complete prohibition.

3. PIC listing of a pesticide is not an international ban

Carbosulfan was included in Annex III of the Rotterdam Convention in 2005 and is therefore subject to the Prior Informed Consent (PIC) procedure. However, PIC listing does not constitute an international ban, nor does it mean that the Rotterdam Convention has determined that carbosulfan must be banned in India. The listing requires importing countries to communicate an informed import decision, which exporting countries must respect. Countries retain the right to determine, under their national regulatory frameworks, whether to continue the registration, use or import of carbosulfan, subject to their obligations under the PIC procedure.

4. Carbosulfan has an important additional role in nematode management

A particularly important consideration in evaluating the consequences of prohibition is that carbosulfan has value beyond its insecticidal activity. Indian research and ICAR technology programmes have documented the use of carbosulfan in integrated management of plant-parasitic nematodes, including root-knot nematodes. Applications have included bare-root dipping of vegetable seedlings with carbosulfan 25 EC and seed treatment with carbosulfan 25 DS, with field evidence of efficacy against root-knot nematodes in crops including tomato, pointed gourd and black pepper.

This aspect deserves particular attention because the chemical nematicide options available to Indian agriculture have already become increasingly restricted. Carbofuran, historically an important chemical nematicide, is now available only under highly restricted formulation- and crop-specific provisions, with Carbofuran 3% CG retained following the 2023 regulatory action. Most recently 474 RC minutes 30.7.2026 granted approval of the registration of Carbosulfan 25% EC for the control of root knot nematode on chilli and cucumber. Consequently, withdrawal of carbosulfan could further narrow an already limited chemical nematicide toolbox and may create an unintended gap in Integrated Pest Management.

5. No unacceptable consumer exposure concern from Carbosulfan

The FSSAI Draft Notification, 2024 proposed carbosulfan MRLs ranging from 0.05 mg/kg to 20 mg/kg, with the highest MRL in dried chilli (20 mg/kg), followed by fresh chilli (2 mg/kg) and rice (0.2 mg/kg). A conservative Theoretical Maximum Daily Intake (TMDI) assessment, using an ADI of 0.01 mg/kg bw/day, 60-kg body weight and GEMS/Food India consumption factors, estimated a total intake of 0.0845 mg/day, equivalent to 14.08% of the ADI. Since the estimated combined exposure based TMDI is only 14.08% of the ADI, the scientific assessment indicates that the proposed carbosulfan MRLs do not pose a chronic dietary safety concern for Indian consumers under the conditions assessed.

6. Alternatives should address both insect and nematode management

Replacement of the insecticidal function of carbosulfan alone would not necessarily constitute complete replacement of its role in plant protection. An alternative insecticide may provide satisfactory control of an insect pest but may have no corresponding activity against plant-parasitic nematodes. Thus, the assessment of alternatives should not be restricted to insect pest control. It should examine crop-wise, pest-wise and nematode-wise replacement packages, particularly in vegetable nurseries, protected cultivation, rice, banana, black pepper and other systems where nematode pressure can be economically important.

Newer nematicides, biological agents, resistant varieties, soil solarisation, organic amendments and other integrated approaches may contribute to replacement. However, their efficacy, registration status, cost, application method and crop coverage differ, and therefore their adequacy should be demonstrated rather than presumed.

7. Agricultural utility and availability of alternatives require crop-specific assessment

Carbosulfan has been registered for economically important crops including rice, chilli, cumin, brinjal and cotton, through different formulations and application methods. Its approved uses encompass several important sucking and chewing pests, including plant hoppers, gall midge, stem borer, leaf folder, aphids, thrips and fruit and shoot borer. No single alternative is demonstrated to reproduce all of these uses across crops. Published Indian studies demonstrate that individual alternatives can provide effective control against particular pests, but these studies should not be interpreted as evidence that a single alternative can replace carbosulfan across its entire registered use profile.

Therefore, before withdrawal, alternatives should be evaluated for:

- biological efficacy at registered rates;
- crop- and pest-specific performance;

- farmer affordability and accessibility;
- residue compliance;
- resistance-management implications;
- environmental and non-target effects; and
- their ability to form part of effective IPM programmes.

8. The scientific basis for the proposed prohibition should be made available

The Draft Order records that an Expert Committee was constituted on 14 January 2026 and submitted its report on 12 June 2026, following which the Registration Committee considered concerns relating to intrinsic hazard, rapid metabolism to carbofuran, toxicity to non-target organisms and international regulatory restrictions before recommending complete prohibition. In this context, I respectfully request that the Expert Committee report and the scientific material placed before the Registration Committee may kindly be made available in the public domain.

This would enable scientists, agricultural institutions and other stakeholders to understand the evidence underlying the change in regulatory position and to provide meaningful, evidence-based comments during the consultation process.

9. Risk mitigation should be evaluated before complete prohibition

The regulatory assessment may also consider whether identified risks can be reduced through strengthened stewardship and targeted risk-management measures, including improved packaging, secure storage, appropriate personal protective equipment, application safeguards, farmer and applicator training, medical preparedness and effective monitoring.

The relevant scientific question is not whether carbosulfan possesses hazard- it clearly does, but whether the magnitude of risk under Indian conditions of approved use is unacceptable and cannot be adequately managed through feasible risk-reduction measures. The background assessment identifies this as a central question requiring evidence on actual exposure and the effectiveness of existing mitigation measures.

10. A scientifically designed transition mechanism is essential if prohibition is pursued

If, after comprehensive assessment, the Government ultimately decides that prohibition is warranted, an abrupt withdrawal could create avoidable pest- and nematode-management gaps. A defined, time-bound transition period would allow crop- and pest-specific as well as crop- and nematode-specific alternatives to be validated, registered and made available to farmers. It would also facilitate orderly management of legally held stocks, consignments already in transit, cancellation of registrations and safe disposal of unusable stocks. Such a transition would not dilute the regulatory objective; rather, it would facilitate orderly implementation while reducing the possibility of unregulated use or unsafe disposal.

11. Collective Recommendation

In our assessment, the regulatory choice should not be reduced simply to one between unrestricted continuation and immediate prohibition. The appropriate objective should be risk reduction through scientifically validated and practically implementable risk-management measures, while ensuring that the withdrawal of an existing tool does not create avoidable

gaps in crop protection. We therefore respectfully recommend that, before finalising the proposed prohibition, the Government may consider:

1. making available the Expert Committee report and the scientific evidence underlying the recommendation for complete prohibition;
2. undertaking an India-specific assessment of human-health risk, including occupational, accidental and intentional poisoning, dietary exposure and residues under Good Agricultural Practices;
3. undertaking an India-specific environmental and ecotoxicological assessment, including exposure of aquatic organisms, pollinators, natural enemies and other non-target organisms under representative field conditions;
4. conducting a crop-wise and pest-wise assessment of alternatives, including efficacy, economics, resistance management and residue/environmental considerations;
5. undertaking a separate crop-wise and nematode-wise assessment of the role of carbosulfan and the availability of registered and technically validated alternatives for plant-parasitic nematodes;
6. evaluating whether strengthened stewardship, restricted use, improved packaging, PPE, application safeguards, training and monitoring can adequately mitigate identified risks where scientifically justified;
7. if prohibition is ultimately considered necessary, providing a defined and time-bound transition period for validation, registration and deployment of effective alternatives, including alternatives for nematode management; and
8. providing for post-regulatory monitoring of farmer costs, pest and nematode-management practices, resistance development, pesticide poisoning and environmental outcomes.

The objective of pesticide regulation should ultimately be measurable risk reduction and safer agriculture while maintaining the resilience, productivity and sustainability of Indian agriculture. In the case of carbosulfan, such an approach is particularly important because its withdrawal has implications not only for insect pest management but also for the already narrowing range of options available for plant-parasitic nematode management. A scientifically robust decision should therefore consider the new evidence underlying the present proposal, the agricultural utility of carbosulfan, its demonstrated nematicidal value, the availability of validated alternatives and the feasibility of risk mitigation under Indian conditions before determining the appropriate regulatory pathway.

India is known for having its independent evaluations and review of pesticides. From 1990 several committees including SN Banerjee (1991), Dr RB Singh (1999), Dr CD Mayee (2006), Dr. Anupam Varma (2013-15), SK Malhotra (2017-18), Dr TP Rajendran (2021-22) were constituted to review the pesticides periodically. These committees headed by expert agricultural scientists used differentiated regulatory outcomes rather than treating all pesticides as requiring the same endpoint. Reviews undertaken by these expert committees have resulted in combinations of continued use with safeguards, further review, restriction and phase-out for different pesticides. Two of these committees where carbosulfan had come up for review, recommended continued use of carbosulfan. Carbosulfan so far didn't come up at all for any discussion before the so called 2026 review.

It is therefore requested that the proposed prohibition may kindly be subjected to a comprehensive, multidisciplinary and India-specific scientific review before the final Order is issued. Furthermore, to ensure that future regulatory decisions are scientifically valid,

practically viable, and closely aligned with ground realities, we strongly urge that the Government of India and the DPPQS make it a mandatory policy to formally include representatives from these apex professional plant protection societies and Plant protection Institutes of ICAR as active members in all crucial technical, review, and pesticide regulatory committees. The exclusion of specialized academic and field-scientific bodies risks creating policy decisions detached from actual agronomic challenges, whereas our formal involvement will guarantee balanced, robust, and scientifically sound recommendations for the nation.

We request you to kindly consider this collective representation from the scientific community in the interest of sustainable agriculture and farmer welfare.

Thanking you,

Yours faithfully,



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Dr. V.V. Ramamurthy
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Copy for information to:

1. The Plant Protection Adviser (PPA), DPPQS, Faridabad.
2. The Secretary, Central Insecticides Board & Registration Committee (CIB&RC), Faridabad.